Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21301

F-401

B.C.A. I Year (First Semester) Examination, 2023-24

(New Course)

Paper - BCA-101

Computer Fundamental and Programming in C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks.

1. (a) Define Computer? Also explain the various features of computer system.

   (b) Explain in detail the history of computer; also explain the evolution process of it.

2. (a) Convert the following Numbers

   (a) $(10101)_2 = (?)_{10}$

   (b) $(1110001)_2 = (?)_{10}$

   (c) $(62)_8 = (?)_{10}$

   (d) $(2085)_8 = (?)_{10}$

   (e) $(1CE)_{16} = (?)_{10}$

   (f) $(2AF)_{16} = (?)_{10}$

   (b) Differentiate between the characteristic of primary and secondary memory of computer.

3. (a) What is Cach. Memory? How it reduce the *mismatch* of processor and main memory speed?

   (b) Define the software. List and explain the types of software. Give two example of each category.

4. (a) Differentiate between High level, Machine & Assembly level language.

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(b) What is Algorithm? define it? Write algorithm for find factorial of a given number? 5

5. (a) Find the position of element 29 using binary search method in an array 'A' given below :

A = {11, 5, 21, 3, 29, 17, 2, 43}

(b) Write an algorithm to delete a node from the beginning of a circular linked list.

6. (a) Explain the storage classes in C with example programs.

(b) Explain the decision making statement in C with example programs.

7. (a) Define Array? Declare a float array of size 5 and assign 5 values to it.

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(b) Write a C program to multiply two matrices (2D array) which will be enterd by a user.

8. (a) Explain Linear search in C.

(b) Explain Bubble sort in detail.

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Paper Code : 21301

F-401

B.C.A. I Year (First Semester) Examination, 2022-23

(New Course)

Paper-BCA-101

COMPUTER FUNDAMENTAL AND PROGRAMMING IN C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions.

1. (a) Differentiate Primary memory with Secondary memory.

   (b) Explain classification of Computer.

2. (a) Explain different types of RAM.

   (b) Explain types of Software.

3. (a) Differentiate ASCII with EBCDIC code.

   (b) Explain the structure of CPU

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4. (a) Explain the advantages & disadvantages of algorithm.

   (b) Explain the steps of program development process.

5. (a) Explain Data types used in C.

   (b) Explain different types of operators used in C.

6. (a) Explain standard C library and C preprocessor.

   (b) Write a C program to print theArmstrong number from 1 to N.

7. (a) Write a C program for bubble sorting.

   (b) Write a C program to sum of two arrays.

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8. (a) Explain with syntax of for and while loop.

(b) Write a C program to print the Fibonacci number from 1 to N.

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Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21301

F-401

B.C.A. (First Semester) Examination, 2021-22

(New Course)

Paper - BCA-101

Computer Fundamental and Programming

Time : Three Hours] ***[Maximum Marks : 70***

Note : Attempt any **five** questions. **All** question carry equal marks.

1. (a) Draw a block diagram of Computer and explain the function of each of the blocks? 8

   (b) Explain the generation of Computer with their features? 6

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2. (a) What are the peripheral devices? Explain with example? 7

   (b) Differentiate between primary and secondary memory on basic of their characteristics? 7

3. (a) Convert the following : 6

   (i) $(17.35)_{10}$ to binary form

   (ii) $(375)_8$ to hexadecimal number

   (iii) $(AB2)_{16}$ to octal number

   (b) What do you mean by fixed and floating point representation? Also differentiate ASCII & EBSDIC code? 8

4. (a) Differentiate among Machine Language Assemble. Language and High level Language. 7

   (b) Define System Software with example. 7

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5. (a) Define Algorithm & Draw a flow chart to find greater between Two numbers? 7

(b) What is structured programming? Discuss about the debugging and testing of programmes? 7

6. (a) Discuss various "C" operators with example. 7

(b) Differentiate "while" and "Do-while" loop with example. 7

7. (a) What do you mean by Sorting? Describe Bubble sort? https://www.mjpruonline.com 7

(b) What are two - dimensional arrays? Write a program in "C" language to add two matrices of 3×3. 7

8. Write short notes on any **four** of the following:

(a) Central Processing Unit 3.5 each

(b) "C" Preprocessor

(c) Linear Array

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(d) Program Correctness

(e) "For" loop

(f) Data types in "C"

(g) Insertion Sorting

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21301

F-401

B.C.A. Ist Year (First Semester)

Examination, 2021

(New Course)

Paper-BCA-101

COMPUTER FUNDAMENTAL AND PROGRAMMING IN C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions.

1. (a) What do you mean by Computers? Discuss various generations of Computer in brief?

   (b) Draw and explain a neat block diagram of Computer?

2. (a) What are various memories, types, explain their characteristics?

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   (b) Write classification of Computer?

3. (a) Convert following :

   (i) $(3AC)_{16} = (?)_2$

   (ii) $(53.72)_{10} = (?)_{16}$

   (iii) $(746)_8 = (?)_2$

   (iv) $(101010111)_2 = (?)_{10}$

   (v) $(5.72 \times 10^2)_{10} = (?)_8$

   (b) What do you mean by ASCII Character?

4. (a) What are various Software types, explain?

   (b) Differentiate between :

   (i) Machine Language and Assembly language

   (ii) Top down and bottem up approach?

5. What are various program planning tools, explain in brief? Draw and explain a flow Chart to find larger of two numbers?

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6. (a) What are arrays? Write a 'C' program to find smallest number of an array of 10 numbers?

   (b) Write a 'C' program to perform searching an element in an array of 10 numbers?

7. (a) What are loops, write a program to print "Hello World" 20 times using any loop?

   (b) What do you mean by debugging explain?

8. Write short notes (any **four**) :

   (i) EBSDIC

   (ii) Library functions

   (iii) Variables in 'C' programs

   (iv) I/O devices

   (v) Testing

   (vi) History of computers

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Paper Code : 21301

F–401

B.C.A. Ist Year (First Semester) Examination, 2018

(New Course)

Paper-BCA-101

COMPUTER FUNDAMENTAL AND PROGRAMMING IN C

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions.

1. (a) Explain the various generations of computers discussing the technical changes and achievement in each generation.

   (b) What do you understand by input-output devices ? Explain *five* input and output devices. 7,7

2. (a) Define various decision control statements in C. If the marks obtained by a student in three different subjects are input through the keyboard. Write a program in C to calculate average marks and percentage obtained by the student.

   (b) Discuss the various types of operators in C. Discuss each with suitable example. 7,7

3. (a) What is Structured programming concept ? Define in detail top down design.

   (b) Discuss flow charts. Also define various flow chart notation used of representation of "C" Programs. 7,7

4. (a) What do you mean by computer software ? Discuss the relationship between software and hardware. Explain different types of software with suitable example of each.

   (b) What is memory ? What are the different types of memories ? Discuss their merits, demerits and area of application. 7,7

5. Define linear array and Multidimensional array with example. Write a program in C to calculate matrix multiplication of two 3 * 3 matrix. 14

6. (a) Covert the following : 2 each

   (i) $(330.625)_{10}$ ..... > $()_2$

   (ii) $(110.1001)_2$ ..... > $()_{10}$

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(iii) $(6782)_8$ ..... > $()_{16}$

(iv) $(9EB)_{16}$ ..... > $()_8$

(v) $(5B7)_{16}$ ..... > $()_{10}$

(b) Explain the following terms : 2 each

(i) ASCII Code

(ii) Bit and Bytes.

7. Differentiate between any *four* of the following : 3.5 each

(i) While loop and Do-while loop

(ii) Multiprocessing and Multiprogramming

(iii) Algorithm and Flowchart

(iv) Break and Continue Statement

(v) Time sharing and Real time processing.

8. Write short notes on any *two* of the following : 7 each

(i) Central Processing unit

(ii) High level language

(iii) Debugging and testing of programs.

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Paper Code : 21301

F–401

B. C. A. (First Semester)
EXAMINATION, 2017-18

(New Course)

Paper No. BCA–101–N

COMPUTER FUNDAMENTAL AND PROGRAMMING

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Define Computer. Explain the characteristics of Computer.

   (b) Define the term 'Memory'. Differentiate between primary and secondary memories.

2. (a) What is ASCII code ? Write a program to convert numbers from 1-50 to its equivalent ASCII value.

   (b) Compare and contrast between Low level language and High level language.

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3. (a) What is Data Representation ? How to represent integers and floating point numbers ?

   (b) Convert the following :

   (i) $(100101)_2$ to $(X)_{10}$

   (ii) $(928)_{10}$ to$(X)_8$

   (iii) $(37.50)_{10}$ to $(X)_2$

   (iv) $(E7B5)_{16}$ to $(X)_{10}$

   (v) $(456)_8$ to $(X)_{10}$

   (vi) $(54398)_{10}$ to $(X)_{16}$

   (vii) $(1001.1010)_2$ to $(X)_{10}$

4. (a) Explain any *five* 'C' operators. Give suitable example for each.

   (b) What is Flowchart ? Discuss the various symbols used in Flowchart. Write an algorithm and flowchart to find maximum number between two numbers.

5. (a) What do you understand by Debugging and testing of programs ?

   (b) Define Preprocessor. Compare and contract Compiler and Interpreter.

6. (a) Describe Control Flow Statements with suitable examples.

   (b) What is Data Type ? Discuss its classification with proper example.

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7. (a) Define Array. Discuss the various operations associated with an array.

   (b) Describe the types of Sorting and Searching with suitable example.

8. (a) Write a program to check whether an entered number is Palindrome or not.

   (b) Write a program to perform addition of two 2 × 2 matrices.

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